

Raju Gadiraju



ABOUT THE ARCHITECT

Enterprise Architect: Healthcare, Finance & AI

Raju Gadiraju

I am Raju, an Enterprise Architect who bridges the gap between executive business strategy and deep, hands-on engineering. My primary focus is transforming healthcare - architecting scalable revenue cycle platforms, administrative workflows, and secure, enterprise-wide clinical systems.

Beyond high-level strategy, I am a highly accomplished engineer specializing in production-grade GenAI systems, sophisticated RAG pipelines, and high-performance multi-cloud infrastructure (AWS, Azure, GCP). By leveraging my extensive background in legacy financial modernization and optimizing large-scale AI training on NVIDIA DGX clusters, I build robust, scalable intelligence platforms that solve mission-critical challenges across both the healthcare, finance and AI.



[Get in Touch](#)

[Read Case Studies](#)

What I Bring

Deep experience across the full enterprise architecture stack - from initial stakeholder alignment through system design, and multi-cloud AI delivery.



Healthcare Systems Architecture

Architecture for revenue cycle, admin, and clinical platforms. Familiarity with healthcare interoperability standards including HL7 / FHIR, EHR/EMR, Revenue Cycle Management (RCM), and HIPAA compliance.



Enterprise Architecture

End-to-end multi-cloud infrastructure design across AWS, Azure, and GCP. Secure, scalable Kubernetes (EKS, AKS) deployments, CI/CD automation, and MLOps lifecycle management.



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Optimization of multi-node NVIDIA DGX (H100/A100) clusters. Deep expertise in TensorRT, Triton, vLLM, and Slurm scheduling to maximize hardware utilization and minimize inference latency.

Strategic implementation of Anthropic's Claude, GPT-4, and open-source models into production. Building high-accuracy RAG pipelines, fine-tuning, and robust evaluation frameworks.



Cloud Agentic AI

Designing autonomous and semi-autonomous multi-agent systems using LangGraph and CrewAI. Implementing hierarchical supervisor-worker architectures and secure enterprise tool integrations.



Custom AI Development

Full-cycle engineering of AI applications. From developing Golang gRPC microservices and Model Context Protocol (MCP) servers to building secure, scalable backend data pipelines.



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leadership on AI roadmaps, vendor selection, infrastructure cost-optimization, and technical debt mitigation in legacy-to-AI modernizations.

Migrating legacy monolithic systems (1000+ COBOL financial systems) to Java/Spring Boot and C# .NET. Utilizing fine-tuned CodeBERT and GraphCodeBERT models for business logic extraction.



Systems Integration

Designing integrations across complex enterprise landscapes. API strategy, middleware, event-driven architectures (Kafka, Airflow), and end-to-end EA frameworks including TOGAF and ArchiMate.

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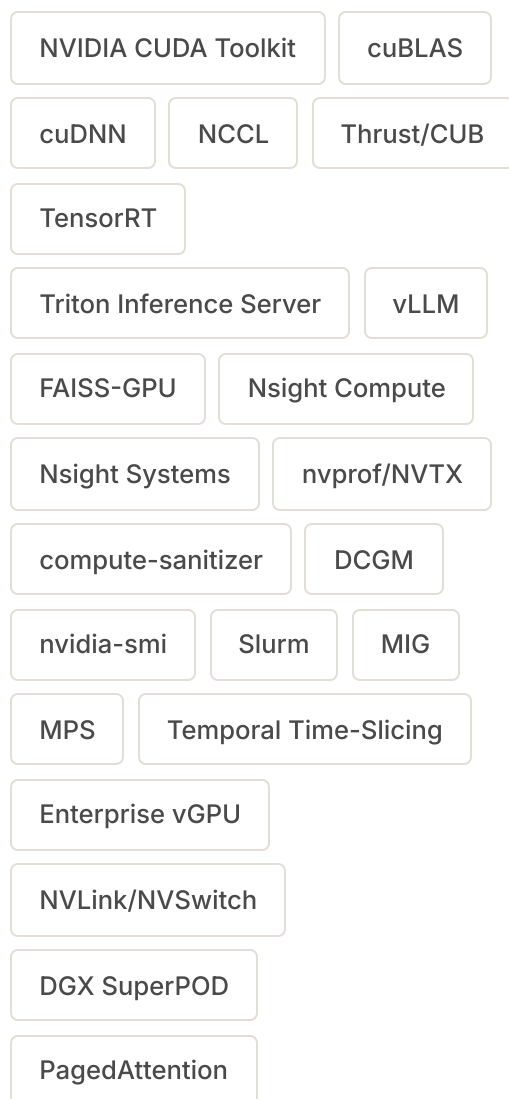


Technical Toolkit

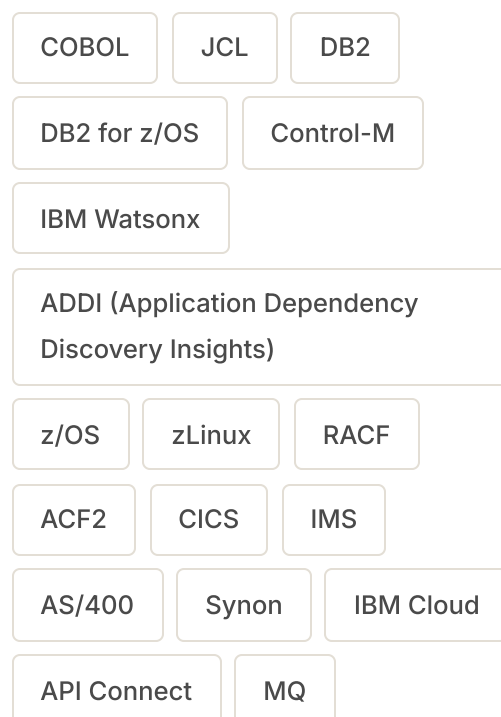
A comprehensive inventory of platforms, languages, and frameworks utilized in production environments.



GPU / HPC



IBM Mainframes & Ecosystem



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NVIDIA GPU Operator

RunAI

NVIDIA Mission Control (BCM)

NGC Container Registry

Spectrum-X Ethernet

BlueField DPUs

GPFS/Lustre

NVIDIA Omniverse

HPE ProLiant AI Compute

HPE iLO

Liquid Cooling



Programming Languages

Golang

Python

C

C++

Java

COBOL

SQL

PL/SQL

SparkSQL

Shell

Bash

Perl

Agentic AI & RAG

LangGraph

LangChain

CrewAI

Semantic Kernel

AutoGen

AWS Bedrock Agents

Azure AI Agent Service

GCP Vertex AI Agent Builder

Model Context Protocol (MCP)

OpenAI Functions

FAISS

Pinecone

ChromaDB

Weaviate

RAGAS

AI/ML Platforms

AWS SageMaker

AWS Bedrock

Azure OpenAI

Azure AI Foundry

Azure AI Agent Service

Azure ML

GCP Vertex AI

Vertex AI Gemini

IBM Watsonx

Hugging Face

OpenAI API

MLflow

Kubeflow

NVIDIA NIM

Weights & Biases

ML Frameworks & Models



PyTorch

TensorFlow

Keras

Hugging Face Transformers

CodeBERT

GraphCodeBERT

CodeT5

Semantic Kernel

AutoGen

ONNX

TensorRT

Ray

Optuna

TFX

DeepSpeed

FSDP

AutoML

XGBoost

LightGBM

CatBoost

LSTM

Seq2Seq

FastText

SentencePiece

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CloudFormation	RDS	EKS	
VPC	NAT	WAF	KMS
ACM	SG	ECR	EBS
EFS	ECS	Fargate	
Bedrock	Bedrock Agents		
Route 53	CodePipeline		
CodeBuild	CodeDeploy		
Direct Connect	SageMaker		
Rekognition	Textract		
Transcribe	Polly		
Comprehend	Lex		
Translate	Athena	Glue	
Redshift	Lambda	Kinesis	
QuickSight	App Mesh		
Step Functions	Inferentia		
Q Developer	AWS Batch		
AWS Autopilot			

GCP

CDN

Azure OpenAI

Azure AI Agent Service

ADF

Azure Databricks

Key Vault

Azure Monitor

Azure DNS

SQL Database

VNet

NAT Gateway

ExpressRoute

Private Endpoint

Service Fabric

ADLS

Functions

Logic Apps

Cosmos DB

Service Bus

Azure ML

Cognitive Services

Container Registry

App Service

API Management

Application Gateway

Load Balancer

Security Center

Azure AutoML

Resource Manager

OCI



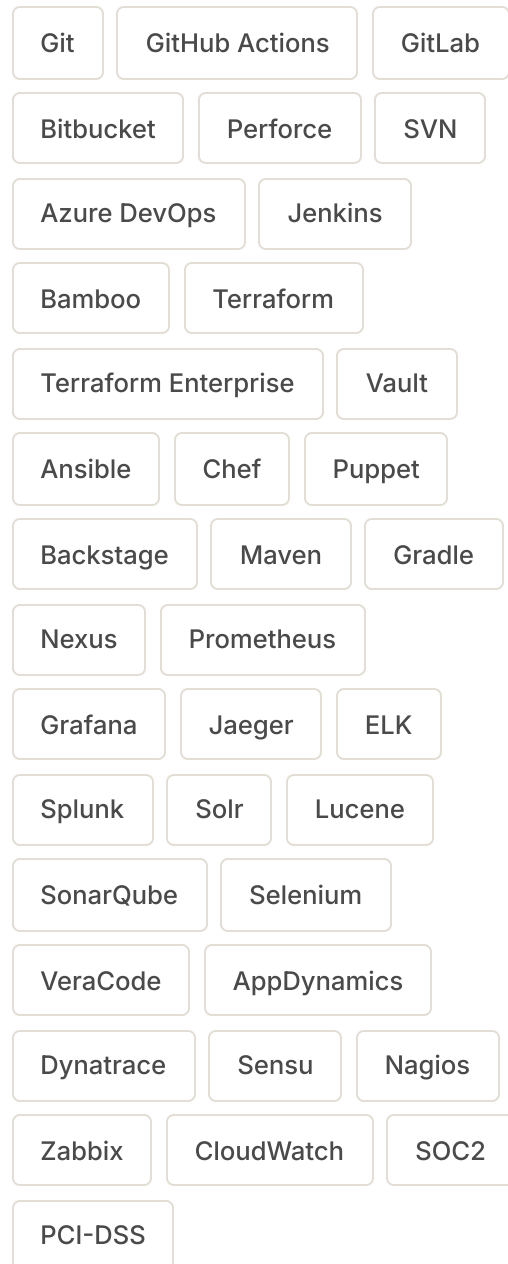
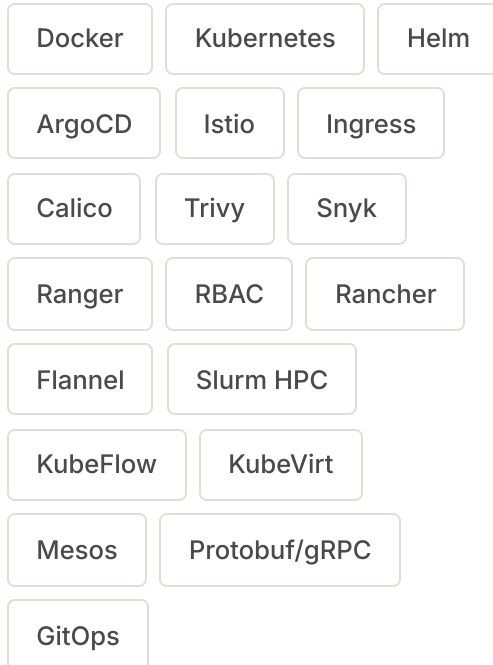
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Looker	Vertex AI	Monitoring & Logging
Vertex AI Agent Builder	Gemini	Database Services
Cloud Build	Cloud Composer	OKE
Cloud Deployment Manager		Data Integration
Anthos	Service Mesh	Oracle Integration Cloud
App Engine	Cloud Functions	Autonomous Data Warehouse
Cloud Run	Pub/Sub	Multicloud Services
Dataflow	Firestore	
Bigtable	Spanner	
Cloud DNS	Cloud KMS	
Secret Manager		
Container Registry	Firebase	
StackDriver		
Cloud Load Balancing		
Cloud Monitoring		
Cloud Logging		
Storage Transfer Service		



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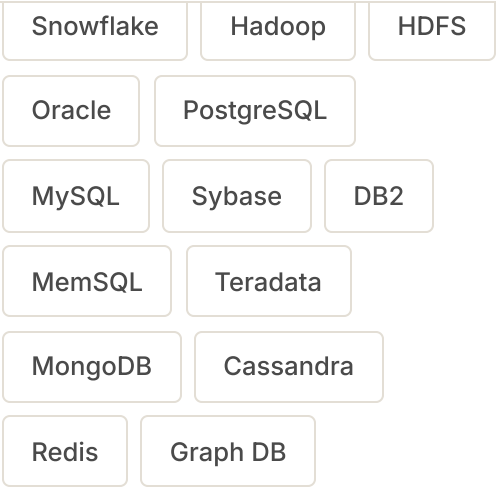
Orchestration



Data & Databases



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Professional Experience



APR 2024 – PRESENT

Software Consultant / Infrastructure Engineer

Ema Unlimited



- Built production RAG pipelines on Azure AI Search and Pinecone using hybrid retrieval and cross-encoder reranking, ensuring high accuracy and sub-second latency.
- Developed sophisticated MCP servers in Golang to connect autonomous agents to Salesforce, ServiceNow, and Jira with secure RBAC and schema versioning.
- Profiled AI workloads across NVIDIA DGX H100/A100 clusters using Nsight Systems and DCGM; built optimized edge AI inference systems using TensorRT.



JUN 2023 – APR 2024

Senior Consultant / Cloud Systems Engineer

Fidelity Investments

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- Built distributed ML pipelines with PyTorch DDP and TensorFlow across multi-node GPU clusters, tracking experiments via MLflow.
- Trained NLP pipelines to extract critical business logic from legacy JCL and Control-M environments for automated mainframe modernization analysis.



AUG 2019 – MAY 2023

Senior Principal Consultant

Genpact Digital & HPE Labs AI ML JV Team

- Integrated HPE ProLiant Compute XD685 (AMD EPYC/Instinct MI325X) for optimized high-density rack deployments and HPC efficiency.
- Built complex aviation risk assessment data pipelines integrating GCP, AWS, Azure, and Oracle Autonomous Database, leveraging Airflow for microservice orchestration.
- Designed and implemented synthetic data pipelines to evaluate AI models under realistic and edge-case conditions, improving model robustness and reliability.



JUL 2018 – JUL 2019

Senior Platform Engineer

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AKS, and Spark, establishing automated CI/CD pipelines and Apache Airflow orchestration for Fund Manager dashboards.

- Configured secure Azure Databricks environments integrating Active Directory (AD DS), Private Link, and Key Vault for enterprise credential management.
- Developed a highly scalable GCP-based conversational AI chatbot utilizing Pub/Sub, Dataflow, Firestore, and BigQuery for real-time analytics.



JUL 2016 – FEB 2018

Principal Infra Developer

Cognizant Technologies

- Engineered an AWS-based high-throughput data platform for HBO Live Stream, utilizing Kafka and containerized microservices to optimize real-time event ingestion.
- Modernized enterprise data pipelines on Microsoft Azure, integrating Databricks, Azure Data Factory, and Informatica ETL for secure, cross-layer data orchestration.



NOV 2010 – JUN 2016

Senior Software Engineer

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- Spearheaded big data integration for the Oracle Fusion Product Family across the Hadoop ecosystem (Hive, HBase, Oozie, ZooKeeper).
- Provisioned and optimized enterprise-scale hardware and software environments, including Oracle Exadata, Exalogic, and Cloudera platforms.
- Managed mission-critical DB2 and DB2 for z/OS database environments, ensuring high availability and seamless integration for Siebel CRM applications.



JUN 2008 – NOV 2010

SQA Engineer

Symantec / Veritas

- Architected High Availability and Disaster Recovery (HA/DR) solutions for Oracle RAC, Sybase, and DB2 clusters across diverse Unix/Linux environments.
- Integrated Veritas Clustered File Systems with enterprise SAN/NAS/NetApp storage volumes for optimal data consistency and reliability.
- Developed comprehensive Python and Shell automation scripts to streamline database deployment, monitoring, and disaster recovery processes.



Case Studies

Architecting systems that scale across Healthcare, Finance, and Enterprise domains.

HEALTHCARE · REVENUE CYCLE SYSTEMS



RCM Platform Architecture for Multi-Site Practice

CONTEXT

Multi-site group

SCALE

8 locations

DURATION

6 months

A growing primary care group was managing claims across disconnected systems, causing reconciliation errors and delayed payments. Designed a unified RCM architecture with a central data model, integrated clearinghouse connectivity, and automated denial workflows.

HEALTHCARE · ADMIN SYSTEMS

EHR Integration & Patient Flow Redesign



CONTEXT

DPC Practice

SCALE

Startup

DURATION

3 months

A new direct primary care practice needed to choose and integrate an EHR with membership payments and patient communication tools. Led vendor evaluation, defined integration architecture, and documented the target-state data flows before any implementation began.

Outcome: Avoided costly mid-implementation pivot; practice launched on time with clean system boundaries.

CONTEXT

Health System

SCALE

Enterprise

DURATION

12 months



Led the architecture workstream for migrating a legacy on-premise admin platform to a cloud-based system. Defined current and target state, managed stakeholder alignment across clinical, finance, and IT, and established integration patterns for 14 connected systems.

Outcome: Zero-downtime cutover; all 14 integrations live on day one.

ENTERPRISE · EARLY STAGE

System Blueprint for Healthcare Startup

CONTEXT

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DURATION

6 weeks

Engaged pre-product to help a healthcare founder think through their technical architecture. Produced a system context diagram, data flow maps, platform selection criteria, and a phased build roadmap aligned to their funding milestones.



Outcome: Architecture document used to align technical co-founder and inform Series A conversations.

FINANCE · LEGACY MODERNIZATION

Financial Systems AI Modernization

CONTEXT

Fidelity Investments

SCALE

1000+ Legacy Systems

DURATION

10 months

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Outcome: Engineered fine-tuned CodeBERT and GraphCodeBERT models. Deployed RAG systems using legacy codebase and financial rules for LLM-guided modernization recommendations with 95% semantic accuracy.

HEALTHCARE · AGENTIC AI & AUTOMATION



Agentic AI Workflow for GPI Reclassification

CONTEXT

Enterprise PBM (OptumRx / RxCLAIM)

SCALE

12,700+ records/cycle

DURATION

Cycle reduced from 8 weeks to 1-2 weeks

Manual GPI reclassification and NDC mapping across RxCLAIM systems took up to 8 weeks and cost \$1.5M annually. Mapping errors caused \$630K in yearly remediation and disrupted care for 65,000+ members via claim denials.

Outcome: Architected a 3-agent Azure AI platform (GPT-4 & Azure ML) to automate ingestion, client segmentation, and batch orchestration. Blended



Research & Publications

Academic and industry contributions advancing Edge AI, RAG optimization, and autonomous architectures.



IEEE PUBLICATION

Design of Real-Time Compute Model

Optimizing Architecture Using RAG and Tiny-BERT Models for high-efficiency, low-latency deployments.

ECPDC 2025 (ACM) December 2025

Edge AI Consensus: Blockchain-Secured Federated Learning for Trustless Intelligence Network

A novel framework combining blockchain technology with federated learning for trustless collaboration of edge devices, employing a reputation-based consensus mechanism to filter malicious contributions.

[Read Paper →](#)

Project Nanda

Architecting the Internet of AI Agents. Led the design and implementation of a Kubernetes-native automation control plane for air-gapped GDC deployments.

Automated Feature

Pipelines for LLM-Based Executive Intelligence Systems in Hybrid Cloud Environments

A framework that integrates automated feature pipelines into LLM-based executive intelligence systems, enabling scalable data organizing, feature selection, and modification for hybrid cloud deployment.



[Read Paper →](#)

IEEE GINOTECH 2025

May 2025

Federated Learning Meets Swarm Intelligence: A Privacy-Centric Framework for Big Data Processing

Blends Federated Learning (FL) and Swarm Intelligence (SI)

IEEE ICOCT 2025

June 2025

Federated-Aware Cluster Computing for Resilient and Scalable Machine Learning

Introduces the FACC framework to align decentralized federated computation with high-performance cluster and grid

tackle privacy, scalability, and real-time processing in big data analytics.

[Read Paper →](#)

[Read Paper →](#)

IEEE ICDICI 2025

July 2025



DataOps Meets LLMOps: Automating Cloud-Based AI Workflows from Data Ingestion to Prompt Optimization

A unified framework that bridges DataOps and LLMOps to automate end-to-end cloud-based AI workflows, from raw data processing to continuous prompt optimization for Large Language Models.

[Read Paper →](#)



Engagement Model

How I work with enterprise organizations and growing startups.



01

Architecture Discovery & Assessment

FIXED SCOPE

Engaging to define current state architecture and assess complex enterprise landscapes before committing to technology.

02

System Blueprint for Startup

FIXED SCOPE

03

Fractional EA Advisory

ONGOING

Ongoing architectural leadership. Defining current and target state, roadmaps, and governance models for complex organizations.



Let's Connect

Available for conversations this week. Whether you're at the idea stage or mid-build, early architecture alignment saves significant time and cost down the road.



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